

Project Alpha: Brand-First Invoice Generator

1. Overview & Objectives

1.1 Problem Statement

Standard invoice generators limit brand identity to an unstyled logo upload. For design-conscious businesses, this creates a visual disconnect at the critical point of transaction, rendering generic financial documents that fail to match their premium brand presentation.

1.2 Target Audience

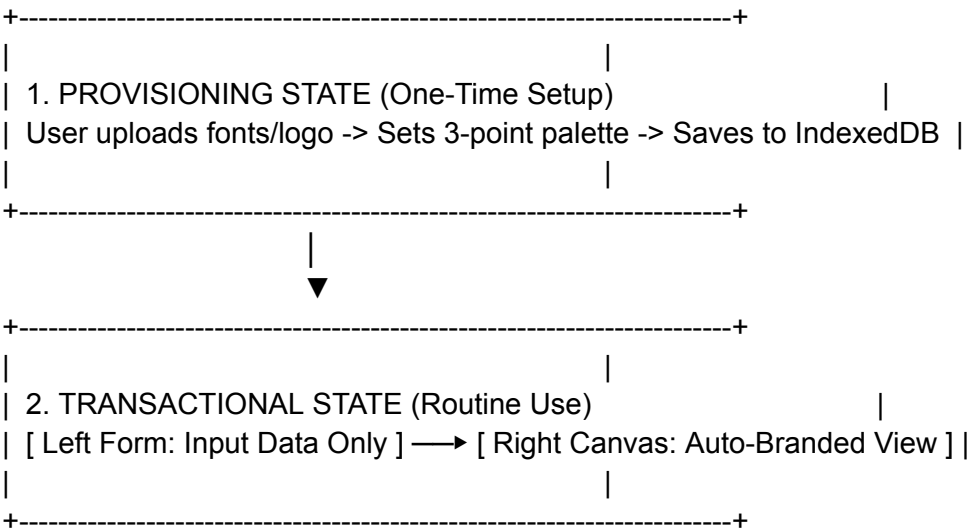
Boutique agencies, freelance creatives, and premium service providers who value visual consistency and require high-fidelity typography, strict semantic brand colors, and proper corporate tax identifiers across all customer-facing touchpoints.

1.3 Value Proposition

A fast, lightweight, split-screen invoice generator where the document design strictly enforces custom typography (via Google Fonts or proprietary uploads), functional color mapping, and crisp vector brand watermarks.

2. User Workflows (Separated States)

The application splits the user experience into two operational modes to ensure that deep brand customization does not add friction to routine invoice creation.



2.1 The Provisioning Track (One-Time Configuration)

1. User opens the **Brand Workspace** settings pane.
2. User uploads their primary brand logo and chooses a typography method (searching Google Fonts or dropping a proprietary font file).

3. User maps three semantic brand colors and uploads a brand watermark asset, adjusting its opacity via a range slider.
4. User fills in their global corporate financial metadata, including default bank account infrastructure (including branch clearing details) and corporate tax identification number (NPWP).
5. User clicks "Save Theme." The engine compiles and serializes the assets into persistent Base64 strings inside the browser's local database.

2.2 The Transactional Track (Routine Generation)

1. User clicks "Create New Invoice."
2. The UI initializes with the master brand styles, bank instructions, and tax metadata already locked and applied to the right-hand preview panel.
3. User utilizes the left-side data form to input client metadata, line items, taxes, and discounts using rapid tab-key navigation.
4. User reviews the live preview and clicks "Export PDF" to immediately download the branded asset.

3. Functional Requirements

3.1 Split-Screen Workspace UI

- **Requirement:** The workspace must be split vertically into two distinct panels.
 - **Left Panel (Data Entry):** Consists of standard HTML form controls (,).
 - **Right Panel (Document Preview):** A read-only, responsive CSS container representing the physical page sheet.
- **Behavior:** As data is typed into the left panel, the right panel must update in real time via reactive state bindings without layout lag or page flashes.

3.2 Advanced Typography Selection Engine

- **Track A (Google Fonts Integration):** The interface must provide a searchable selection of curated Google Fonts. Selection must dynamically inject a font stylesheet link into the DOM head.
- **Track B (Proprietary Font Upload):** Users can upload local font assets strictly limited to .ttf, .otf, and .woff2 formats.
- **Behavior:** The application must read the file via a JavaScript FileReader entirely in-browser, converting it into a Base64 data URI and injecting it into a dynamic @font-face CSS block assigned to the document container.

3.3 Three-Point Semantic Color Mapping

The configuration must map a cohesive brand palette across the invoice structure rather than using a single, disruptive accent color:

- **Primary Accent:** Applied to structural header blocks, horizontal table dividers, or total highlights.

- **Header Text:** A deep neutral color specifically designated for primary titles, metadata labels, and company identifiers to guarantee legal legibility.
- **Muted Background Fill:** A soft neutral or tint used for alternating table rows or the background container of payment terms.

3.4 Positioned Brand Watermark

- **Requirement:** The system must accept an image file upload (.png, .svg) to act as a background asset behind the central invoice text.
- **Constraints:** The watermark module must expose a native range slider mapping directly to the element's CSS opacity property, strictly bounded between 0.03 and 0.12 to ensure the asset never interferes with text readability.

3.5 Core Accounting & Financial Architecture

- **Currency Standard:** The application is locked exclusively to Indonesian Rupiah (IDR).
- **Formatting Rules:** All financial figures must format natively to the standard Indonesian layout: prefixing the currency symbol with a space separator, using periods as thousands separators, and omitting decimal cent values (e.g., Rp 15.000.000).
- **Subtotal Adjustments:** The left panel data form must provide toggleable modifiers that append structural blocks to the bottom summary card of the right preview:
 - *Tax Field:* Percentage-based input (e.g., PPN 11%) that calculates based on the subtotal.
 - *Discount Field:* A flat IDR deduction or a percentage deduction.
 - *Shipping/Other Fees:* A flat numeric IDR addition.

3.6 Localized Corporate Metadata & Compliance Elements

- **Structured Bank Transfer Card:** The configuration forms must expose four dedicated text input fields to ensure structured payment formatting and compliant B2B corporate clearing in the invoice footer:
 - *Nama Bank* (e.g., Bank Central Asia)
 - *Kantor Cabang / Branch Location* (e.g., KCP Sudirman Jakarta)
 - *Nomor Rekening* (Account Number)
 - *Atas Nama / AN* (Account Holder Name)
- **Tax Registration Metadata (NPWP):** The interface must include an input field dedicated to the business's *Nomor Pokok Wajib Pajak* (NPWP).
 - *Formatting:* The field must accept the standard Indonesian tax identification format.
 - *Placement:* When populated, the NPWP must render clearly within the primary company metadata section at the top of the invoice preview container alongside the physical company address.
- **Fiscal Meterai Anchor:**
 - *Trigger:* When the invoice total amount surpasses **Rp 5.000.000**, the right preview panel must automatically render a structural $\text{\$3 \times 3\text{ cm}}$ dashed boundary layout block in the footer next to the signature line.

- *Visuals:* The box must display a muted placeholder instruction text: *"Tempel Meterai Rp 10.000 di sini"*. If the invoice total is beneath the threshold, this block remains hidden unless manually toggled on via the left panel options.

4. Non-Functional & Technical Requirements

4.1 Browser Storage & Data Retention

- **Storage Engine:** Due to the strict 5MB cap of standard localStorage, the application must utilize **IndexedDB** (via a lightweight wrapper like localForage) to persist serialized Base64 custom font structures, logo assets, bank details, and NPWP settings.
- **Data Retention:** To maximize privacy and eliminate server costs, all variables, inputted transaction data, and assets exist purely inside the local browser database. No user session data is transmitted to an external backend server.
- **File Size Bounds:** Custom font file uploads must be strictly capped at a maximum of 4MB to prevent system memory degradation during client-side PDF generation.

4.2 High-Fidelity PDF Generation Pipeline

- **Font-Load Safeguard:** The application must utilize the browser's native FontFaceSet API (document.fonts.ready) to block the compilation engine until all custom external or Base64 fonts are verified as fully loaded in memory. This eliminates the risk of unstyled fallback text rendering in the output document.
- **Vector Fidelity:** The compilation engine must preserve crisp vector text output rather than degrading numbers and addresses into a rasterized screenshot image. Elements tagged with a .no-print class (such as interactive UI buttons) must be automatically stripped from the final output frame during compilation.

4.3 Pagination & Layout Defenses

- **Multi-Page Overflow:** The preview container on the right must visually segment into explicit multi-page sheets if the quantity of line items overflows a single physical page boundaries.
- **Print Layout Integrity:** The rendering pipeline must enforce strict CSS print rules:
 - page-break-inside: avoid; applied to table rows to prevent a single line item description from clipping horizontally across pages.
 - Automatic duplication of the primary brand header text or watermark transparency rules onto subsequent pages if an overflow event occurs.

4.4 Mathematical Precision & Form Guards

- **Calculation Engine:** The application must avoid native floating-point math errors by converting all currency values into whole integer counts during internal subtotal accumulation before applying formatting layers.
- **Form Validation:** The "Export PDF" trigger must remain disabled until required transactional inputs are filled on the left panel (Invoice Number, Client Entity Name, Bank Details, and at least one active Line Item).

- **Default Fallbacks:** If a user bypasses the One-Time Setup track, the system must initialize the right-hand panel with a clean, professional typography stack (e.g., Inter font) and neutral charcoal color rules so the app functions instantly out of the box.

5. Appendix: Engineering Test Scenarios

To ensure absolute layout and engine stability, the development team must validate code releases against the following three threshold configurations:

- **Scenario A (The Enterprise Agency File Weight):** System must process a 3.8MB custom corporate .otf font file, a high-resolution logo, an active 15-digit formatted NPWP string, and a \$2000 \times 2000\text{ px}\$ background watermark setting at 6% opacity. The client-side memory loop must render and compile the vector PDF file in under 2.5 seconds without causing browser execution lag.
- **Scenario B (The Narrative Multi-Page Description):** Input a project row containing a line-item description block exceeding 600 characters of text. The rendering library must cleanly route the expanding text block across the page boundary onto page two, maintaining exact text character alignment without slicing horizontal text rows in half.
- **Scenario C (The Corporate Scale Value):** Input a transaction item volume totaling billions of Rupiah (e.g., subtotaling Rp 2.750.000.000 with an 11% PPN calculation block active). The numbers table on the preview canvas must mathematically render the periods as thousands separators flawlessly without wrapping the text numeric string onto a secondary layout line, keeping right-aligned ledger margins unbroken.